

Project Pitch: TeleCare Triage Tracker

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Abstract

TeleCare Triage is a streamlined web solution designed to improve how patients and providers make safe, efficient care decisions. The project introduces a minimum viable product (MVP) built on a Django API, a PostgreSQL database, a rule-based triage engine, and a curated rules library sourced from trusted clinical guidelines. A reporting worker exports activity summaries in CSV and PDF formats, helping teams evaluate usage and accuracy. This companion document outlines the problem context, stakeholders, scope, MVP components, success metrics, and iterative design strategy. It also presents a clear system sketch, a risk register, and an evidence base with authoritative references.

Problem

As healthcare organizations expand virtual and in-person access, they face a growing challenge: helping patients and caregivers choose the safest, most efficient point of care. Without structured guidance, individuals may delay attention to serious conditions or overwhelm providers with minor concerns. Existing nurse triage lines are valuable but often strained, and their documentation varies in depth and clarity. Although evidence-based resources from AHRQ, CDC, and the ATA provide strong clinical guidance, they remain siloed from lightweight tools that front-line staff can use in real time. TeleCare Triage responds to this gap by combining a rule-driven engine, a curated library of standards, and simple reporting features, giving organizations a scalable way to improve accuracy, speed, and transparency in care-routing decisions.

Stakeholders

- **Patients and caregivers:** Rely on the tool to understand safe next steps and avoid unnecessary visits or delays in urgent care.
- **Nurses and triage staff:** Use the system to document symptoms consistently, apply evidence-based rules, and prioritize cases quickly.
- **Physicians and advanced practitioners:** Benefit from clearer intake summaries and organized queues, allowing them to focus on care delivery instead of manual filtering.
- **Clinic managers and quality teams:** Monitor triage accuracy, turnaround times, and adherence to standards, supporting ongoing improvement and training.

Scope

In-scope for the Minimal Viable Artifact (MVA)

- **Interactive web interface** for entering patient symptoms and context.
- **Rule-based triage engine** using guidance from AHRQ, CDC, and ATA to classify cases as urgent, routine, or self-care.
- **PostgreSQL database** to store encounters, rules, and audit history.
- **Lightweight queue** for staff to view, filter, and act on pending triage items.
- **Reporting service** that exports daily or ad-hoc CSV/PDF summaries for supervisors and quality teams.

Out-of-scope (for the minimal viable artifact)

- Machine-learning predictions or automated diagnosis beyond explicit rules.
- Full EHR or telehealth platform integrations (FHIR, video visits, billing).
- Automated scheduling or write-back to patient records.
- Advanced analytics or dashboards beyond basic exports.

Success Metrics

- **Routing accuracy:** $\geq 90\%$ of encounters correctly directed to the appropriate level of care (self-care, nurse callback, tele-visit, or in-person) based on chart review.
- **Response efficiency:** Median time from patient submission to first provider action ≤ 5 minutes for urgent cases.
- **Safety proxy:** All “red-flag” symptoms escalated according to published guidelines with no unreviewed high-risk items in the queue.
- **User satisfaction:** $\geq 80\%$ positive rating (CSAT/NPS) from patients and staff during pilot use.
- **Adoption:** $\geq 70\%$ of telehealth-related patient contacts documented through the tool during the evaluation period.

Minimal Viable Artifact (MVA)

The MVP delivers three core components:

1. **Triage Intake Form:** A simple web interface where patients or staff can log symptoms, onset, and urgency using structured fields and guided prompts.
2. **Rules Engine & Library:** A lightweight classifier powered by a curated set of triage rules from AHRQ, CDC, and ATA, producing urgent/routine/self-care recommendations with clear rationale.
3. **Queue & Reporting Module:** A streamlined provider view listing active cases, with basic filters, status updates, and an export function for CSV/PDF summaries.

These pieces create a usable baseline that enables accurate routing, quick provider follow-up, and measurable audit data—while leaving advanced analytics or telehealth integrations for later iterations.

Iterative Design Approach

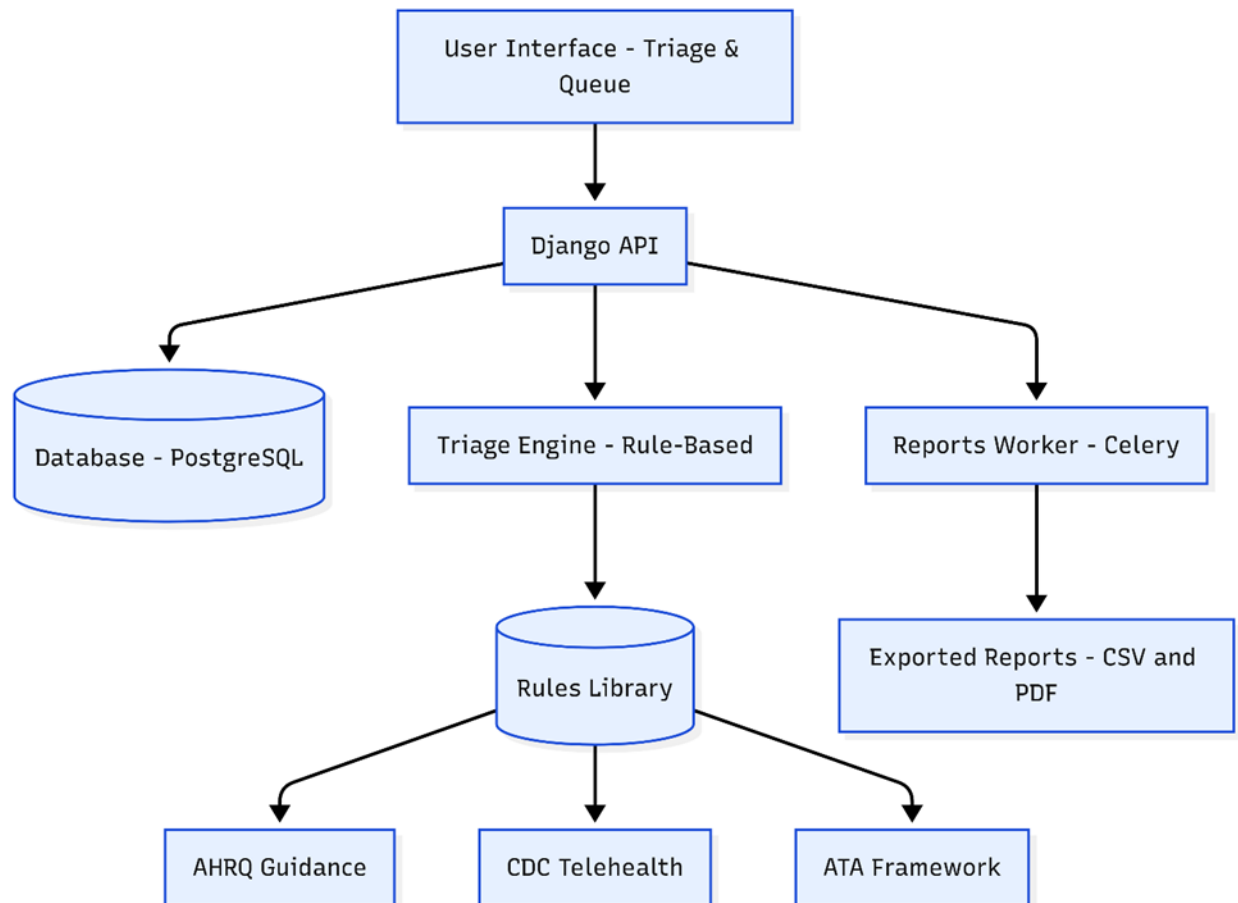
- Development will follow a repeatable cycle:

Planning & requirements → Analysis & design → Implementation → Testing → Evaluation & review

- **Cycle 1 – Core logic and data model:** Build the intake form, rule engine, and database schema. Validate logic against AHRQ/CDC/ATA guidelines and run a limited pilot.
- **Cycle 2 – Queue & usability:** Add the provider queue with filters and status tracking; refine the interface based on feedback about speed and clarity.
- **Cycle 3 – Reporting & rule tuning:** Enable CSV/PDF exports for audits and adjust rules from pilot data, ensuring escalation logic meets safety targets.

Each cycle ends with a short review against routing accuracy, timeliness, and adoption metrics. This keeps the MVP bounded and transparent while still allowing improvements based on user experience.

System Sketch



Evidence Base

- AHRQ (2025). Telephone Triage and Advice Protocols. <https://www.ahrq.gov/>
- CDC (2023). Telehealth Practice Among Health Centers During COVID-19 Pandemic. <https://www.cdc.gov/mmwr/volumes/69/wr/mm6950a4.htm>
- American Telemedicine Association (ATA) (2024). Practice Guidelines for Telehealth and Remote Patient Monitoring. <https://www.americantelemed.org/> (“see Resource Library for current guideline PDF”)
- HHS Telehealth (2024). Telehealth for Providers. <https://telehealth.hhs.gov/providers>

- ONC – Office of the National Coordinator for Health IT (2025). Patient Safety and Health IT: Safe Design and Implementation. <https://www.healthit.gov/topic/safety>

Risk Register

Risk	Impact	Likelihood	Mitigation
Incorrect routing of urgent cases	High	Medium	Use conservative rule thresholds, allow human override, and conduct periodic audits of triage outcomes.
Low adoption by staff	Medium	Medium	Provide targeted training, align tool with existing workflow, and gather early feedback to refine usability.
Scope creep beyond MVP	Medium	High	Keep an explicit backlog; defer advanced features until after initial evaluation; maintain a clear change-control process.

Academic Integrity & AI Disclosure

I used ChatGPT for formatting, minor wording suggestions, and Mermaid for the system sketch layout. I also consulted the instructor's *Module 1 Project Pitch Companion* for guidance on structure and formatting. All design decisions, analysis, evidence selection, and final conclusions presented here are my own.